

第四章 量子力学的矩阵形式与表象理论

§4.0 绪论

一、表象

- 量子力学中描写量子态和力学量的方式不是唯一的
- 一种具体的表达方式称为一种表象
- 量子力学的规律和所用的表象无关, 选用何种表象要依便于所讨论的问题而定.

【思考1】 在一种新表象中，量子态如何表达？

【思考2】 在一种新表象中，力学量如何表达？

【思考3】 在不同表象中，量子态和力学量的表达如何相互转化？

【思考4】 在一种新表象中，量子力学的规律如何表达？

二、量子力学的理论形式

- Hensenberg、Born、Jodan的矩阵力学
线性代数方法, 赋予每个物理量以一个矩阵.
- Schrödinger波动力学(1926年初)
波动方程: 波动函数对空间坐标的二阶微商的偏微分方程
- Dirac和Jordan提出变换理论的普遍形式, 指出矩阵力学和波动力学只不过是量子力学规律的无限多种表达形式中的两种, 完全等价.
- Feynman 路径积分(20世纪40年代)
突出优点: 易于推广到相对论情况, 因而在量子场论中广泛的应用.

三、矩阵常识

1. 矩阵的乘法

$$C = AB, \quad C_{mn} = \sum_l A_{ml} B_{ln}$$

【注意】矩阵乘法一般不满足交换律, 即 $AB \neq BA$.

如果 $AB = BA$, 则称二矩阵对易.

2. 对角矩阵和单位矩阵

对角矩阵 $A_{mn} = A_m \delta_{mn}$

单位矩阵 $I_{mn} = \delta_{mn}$

3. 转置矩阵 行和列互相调换 $\tilde{A}_{mn} = A_{nm}$

4. 共轭矩阵

矩阵 A 的共轭矩阵 $A^\dagger$ 定义为: 将 A 转置、矩阵元取复共轭

$$(A^\dagger)_{mn} = (\tilde{A})_{mn}^* = (A)_{nm}^*$$

$$(AB)^\dagger = B^\dagger A^\dagger$$

5. 厄米矩阵

$$\mathbf{A} = \mathbf{A}^\dagger$$

$$\mathbf{A}_{mn} = (\mathbf{A})_{nm}^*, \quad \mathbf{A}_{nn} = (\mathbf{A})_{nn}^*$$

厄米矩阵的对角元为实数, 非对角元关于主对角线复共轭反射对称.

6. 逆矩阵 A^{-1}

$$A^{-1}A = AA^{-1} = I$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

非奇异矩阵: 有逆矩阵的矩阵

奇异矩阵: 没有逆矩阵的矩阵

7. 矩阵的迹 矩阵的对角元之和 $SpA = \sum_n A_{nn}$

$$Sp(AB \cdots CDE) = Sp(EAB \cdots CD)$$

$$= Sp(DEAB \cdots C)$$

$$= \cdots$$

§4.1 态和力学量的表象

一、态的表象及矩阵表示

● **坐标表象** $\Psi(\vec{r}, t)$

● **动量表象** $c(p, t)$

$$\Psi(x, t) = \int c(p, t) \varphi_p(x) dp, \quad \varphi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{i}{\hbar} px\right)$$

$$c(p, t) = \int \varphi_p^*(x) \Psi(x, t) dx, \quad \varphi_p^*(x) = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{i}{\hbar} px\right)$$

$$\int |\Psi(x, t)|^2 dx = \int |c(p, t)|^2 dp = 1$$

● Q 表象(离散表象)—态的矩阵表示

力学量 $\hat{Q}$ $\hat{Q}\mu_i = q_i\mu_i, (i=1,2,\cdots)$

本征值集 $\{q_1, q_2, \cdots\}$

本征函数系 $\{u_1(x), u_2(x), \cdots\}$

$$\Psi(x, t) = \sum_n a_n(t) u_n(x), \quad a_n(t) = \int u_n^*(x) \Psi(x, t) dx$$

$\{a_n(t), n=1,2,\cdots\}$ 称为 $\hat{Q}$ 表象中的 “波函数”

$|a_n|^2$: 在 $\psi(x, t)$ 所描写的态中测量力学量 $\hat{Q}$ 所得结果为 q_n 的概率.

$$\Psi(t) = \begin{pmatrix} a_1(t) \\ a_2(t) \\ \vdots \\ a_n(t) \\ \vdots \end{pmatrix}, \quad \Psi^\dagger(t) = (a_1^*(t), a_2^*(t), \dots, a_n^*(t), \dots)$$



$\hat{Q}$ 表象中的态矢量



Hermite共轭的态矢量

$$\Psi^\dagger(t)\Psi(t) = \sum_n |a_n(t)|^2 = \int \Psi^*(x,t)\Psi(x,t)dx$$

● 连续谱和多自由度情形

$$\Psi(x, t) = \sum_n a_n(t) u_n(x) + \int a_q(t) u_q(x) dq$$

$$a_n(t) = \int u_n^*(x) \Psi(x, t) dx, \quad a_q(t) = \int u_q^*(x) \Psi(x, t) dx$$

$$\sum_n a_n^*(t) a_n(t) + \int a_q^*(t) a_q(t) dq = 1$$

$$\Psi(t) = \begin{pmatrix} a_1(t) \\ a_2(t) \\ \vdots \\ a_n(t) \\ \vdots \\ a_q(t) \end{pmatrix},$$

$$\Psi^\dagger(t) = (a_1^*(t), \quad a_2^*(t), \quad \cdots, \quad a_n^*(t), \quad \cdots, a_q^*(t))$$

$$\Psi^\dagger \Psi = 1$$

- 坐标的本征函数在自身表象中的表示

$$x\delta(x-x') = x'\delta(x-x') \Rightarrow x\delta(x-x') = x'\delta(x-x')$$

- 动量的本征函数在自身表象中的表示

具有动量为 p' 的自由粒子的状态

$$\Psi_{p'}(x, t) = \frac{1}{(2\pi\hbar)^{1/2}} e^{\frac{i}{\hbar}p'x} e^{-\frac{i}{\hbar}E_{p'}t} = \psi_{p'}(x) e^{-\frac{i}{\hbar}E_{p'}t}$$

$$c(p, t) = \int \psi_p^*(x) \psi_{p'}(x) e^{-\frac{i}{\hbar}E_{p'}t} dx = e^{-\frac{i}{\hbar}E_{p'}t} \delta(p - p')$$

$$c(p) = \delta(p - p')$$

本征方程 $\hat{p}\delta(p - p') = p'\delta(p - p')$ — 动量表象

$\hat{p}\Psi(x, t) = p\Psi(x, t)$ — 坐标表象

二、算符的矩阵表示

坐标表象

$$\hat{F}\left(x, -i\hbar \frac{\partial}{\partial x}\right)$$

$$\Phi(x, t) = \hat{F}\left(x, -i\hbar \frac{\partial}{\partial x}\right) \Psi(x, t)$$

$\hat{Q}$ 表象

$$\Psi(x, t) = \sum_n a_n(t) u_n(x), \quad \Phi(x, t) = \sum_n b_n(t) u_n(x)$$

$$\sum_n b_n(t) u_n(x) = \sum_n a_n(t) \hat{F}\left(x, -i\hbar \frac{\partial}{\partial x}\right) u_n(x)$$

左乘 $u_m^*(x)$ 并积分

$$\sum_n b_n(t) \int u_m^*(x) u_n(x) dx = \sum_n a_n(t) \int u_m^*(x) \hat{F} u_n(x) dx$$

$$b_m(t) = \sum_n \int u_m^*(x) \hat{F}\left(x, -i\hbar \frac{\partial}{\partial x}\right) u_n(x) dx a_n(t)$$

$$F_{mn} = \int u_m^*(x) \hat{F} \left(x, -i\hbar \frac{\partial}{\partial x} \right) u_n(x) dx$$

$$b_m(t) = \sum_n F_{mn} a_n(t)$$

$$\begin{pmatrix} b_1(t) \\ b_2(t) \\ \vdots \end{pmatrix} = \begin{pmatrix} F_{11} & F_{12} & \cdots & F_{1m} & \cdots \\ F_{21} & F_{22} & \cdots & F_{2m} & \cdots \\ \vdots & \vdots & & \ddots & \end{pmatrix} \begin{pmatrix} a_1(t) \\ a_2(t) \\ \vdots \end{pmatrix}$$

$$F = \begin{pmatrix} F_{11} & F_{12} & \cdots & F_{1m} & \cdots \\ F_{21} & F_{22} & \cdots & F_{2m} & \cdots \\ \vdots & \vdots & & \ddots & \end{pmatrix}$$

在离散表象中算符用(方)矩阵代表

$$\Phi(t) = F\Psi(t)$$

● 表示力学量F的矩阵是厄米矩阵

$$F_{mn} = F_{nm}^* = (F^\dagger)_{mn} \quad (\text{矩阵转置取复数共轭不变})$$

【证明】

$$\begin{aligned} F_{mn} &= \int u_m^*(x) \hat{F} u_n(x) dx \\ &= \int [\hat{F} u_m(x)]^* u_n(x) dx \\ &= \left[\int (\hat{F} u_m(x)) u_n^*(x) dx \right]^* \\ &= \left[\int u_n^*(x) \hat{F} u_m(x) dx \right]^* \\ &= F_{nm}^* = (F^\dagger)_{mn} \end{aligned}$$

【注】 Hermite 算符的定义 $\int \psi^* \hat{F} \phi dx = \int (\hat{F} \psi)^* \phi dx$

$$F^\dagger = F \quad [(F^\dagger)_{mn} = (F_{nm})^*]$$

● 力学量在自身表象中为对角的厄米矩阵

【证明】

$$F_{mn} = \int u_m^*(x) \hat{F} u_n(x) dx$$

$F \Rightarrow Q$

$$Q_{mn} = \int u_m^*(x) \hat{Q} u_n(x) dx = \int u_m^*(x) Q_n u_n(x) dx$$

$$= Q_n \int u_m^*(x) u_n(x) dx = Q_n \delta_{mn}$$

$$= \begin{cases} Q_n, & m = n \\ 0, & m \neq n \end{cases}$$

$$Q = \begin{pmatrix} Q_1 & 0 & 0 & \cdots & 0 \\ 0 & Q_2 & 0 & \cdots & 0 \\ 0 & 0 & Q_3 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix}$$

【结论】 一个算符在其自身的表象中是对角矩阵, 各对角元素就是各本征值.

- 本征值构成既有分立谱又有连续谱

$$Q_1, \quad Q_2, \quad \cdots, \quad Q_n, \quad \cdots, \quad q$$

$$u_1(x), u_2(x), \cdots, u_n(x), \cdots, u_q(x)$$

$$F_{nm} = \int u_n^*(x) \hat{F} u_m(x) dx, \quad F_{nq} = \int u_n^*(x) \hat{F} u_q(x) dx$$

$$F_{qm} = \int u_q^*(x) \hat{F} u_m(x) dx, \quad F_{qq'} = \int u_q^*(x) \hat{F} u_{q'}(x) dx$$

矩阵形式

$$F = \begin{pmatrix} F_{nm} & \vdots & F_{nq} \\ \cdots & \cdots & \cdots \\ F_{qm} & \vdots & F_{qq'} \end{pmatrix}$$

$$Q = \begin{pmatrix} Q_1 & 0 & 0 & \cdots & 0 \\ 0 & Q_2 & 0 & \cdots & 0 \\ 0 & 0 & Q_3 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & q \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix}$$

【例题1】坐标表象中算符的矩阵元

$$\begin{aligned} F_{x'x} &= \int \psi_{x'}^*(x'') \hat{F} \left(x'', \frac{\hbar}{i} \frac{\partial}{\partial x''} \right) \psi_x(x'') dx'' \\ &= \int \delta(x' - x'') \hat{F} \left(x'', \frac{\hbar}{i} \frac{\partial}{\partial x''} \right) \delta(x - x'') dx'' \\ &= \hat{F} \left(x', \frac{\hbar}{i} \frac{\partial}{\partial x'} \right) \delta(x - x') \end{aligned}$$

【例题2】动量表象中算符的矩阵元

$$F_{p'p} = \int \psi_{p'}^* \hat{F} \psi_p dx$$

【例题3】一粒子在一维无限深势阱中运动, 状态为

$$\psi(x) = \frac{4}{\sqrt{a}} \cos^2 \frac{\pi x}{a} \sin \frac{\pi}{a} x$$

求该态在能量表象中的矩阵表示?

【解答】一维无限深势阱中运动的粒子, 其能量算符的本征函数为

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x, \quad n = 1, 2, 3, \dots$$

$$\psi(x) = \sum_n a_n \psi_n(x)$$

$$a_n = \int \psi_n^*(x) \psi(x) dx = \frac{1}{\sqrt{2}} \delta_{1n} + \frac{1}{\sqrt{2}} \delta_{3n}$$

$$a_1 = \frac{1}{\sqrt{2}}, a_3 = \frac{1}{\sqrt{2}}, a_n = 0 \quad (n \neq 1, 3)$$



$$\psi = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \\ \vdots \\ 0 \\ \vdots \end{pmatrix}$$

【例题4】在坐标表象证明 x, p 为厄米算符

【证明】

$$x_{xx'} = x' \delta(x - x')$$

$$x_{xx'}^+ = \left(x_{x'x}\right)^* = \left(x \delta(x' - x)\right)^* = x \delta(x' - x) = x' \delta(x - x') = x_{xx'}$$

$$p_{xx'} = -i\hbar \frac{\partial}{\partial x} \delta(x - x') = -i\hbar \frac{\partial}{\partial (x - x')} \delta(x - x')$$

$$\begin{aligned} p_{xx'}^+ &= \left(p_{x'x}\right)^* = \left[-i\hbar \frac{\partial}{\partial x'} \delta(x' - x)\right]^* \\ &= i\hbar \frac{\partial}{\partial x'} \delta(x' - x) = i\hbar \frac{\partial}{\partial x'} \delta(x - x') \\ &= -i\hbar \frac{\partial}{\partial (x - x')} \delta(x - x') = p_{xx'} \end{aligned}$$

【例5】求一维谐振子的坐标 x 、动量 p 以及Hamilton量 H 在能量表象中的表示.

【分析】 不同体系的Hamilton量不一样, 能量表象的基矢也不一样. 这里能量表象的基矢为一维谐振子Hamilton量的本征函数 $\psi_n(x)$

【解答】 利用一维谐振子波函数的递推关系

$$x\psi_n = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} \psi_{n-1} + \sqrt{\frac{n+1}{2}} \psi_{n+1} \right]$$

$$\frac{d}{dx} \psi_n = \alpha \left[\sqrt{\frac{n}{2}} \psi_{n-1} - \sqrt{\frac{n+1}{2}} \psi_{n+1} \right]$$

可以算出

$$x_{mn} = (\psi_m, x\psi_n) = \frac{1}{\alpha} \left[\sqrt{\frac{n+1}{2}} \delta_{m,n+1} + \sqrt{\frac{n}{2}} \delta_{m,n-1} \right]$$

$$p_{mn} = (\psi_m, -i\hbar \frac{d}{dx} \psi_n) = i\hbar \alpha \left[\sqrt{\frac{n+1}{2}} \delta_{m,n+1} - \sqrt{\frac{n}{2}} \delta_{m,n-1} \right]$$

注意： 这里的 m 、 n 都是由0开始取值.

$$(x_{mn}) = \frac{1}{\alpha} \begin{bmatrix} 0 & 1/\sqrt{2} & 0 & 0 & \dots \\ 1/\sqrt{2} & 0 & \sqrt{2/2} & 0 & \dots \\ 0 & \sqrt{2/2} & 0 & \sqrt{3/2} & \dots \\ 0 & 0 & \sqrt{3/2} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$(p_{mn}) = i\hbar\alpha \begin{bmatrix} 0 & -1/\sqrt{2} & 0 & 0 & \dots \\ 1/\sqrt{2} & 0 & -\sqrt{2/2} & 0 & \dots \\ 0 & \sqrt{2/2} & 0 & -\sqrt{3/2} & \dots \\ 0 & 0 & \sqrt{3/2} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$H_{mn} = (\psi_m, \hat{H} \psi_n) = E_n \delta_{mn} = (n + \frac{1}{2}) \hbar \omega \delta_{mn}$$

$$(H_{mn}) = \hbar \omega \begin{pmatrix} 1/2 & 0 & 0 & 0 & \dots \\ 0 & 3/2 & 0 & 0 & \dots \\ 0 & 0 & 5/2 & 0 & \dots \\ 0 & 0 & 0 & 7/2 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$



对角矩阵

任何力学量在自身表象中的表示都是对角矩阵

三、表象变换 (*)

- 量子态的变换

F 表象(基矢 ψ_k , 假定是离散谱) $(\psi_k, \psi_j) = \delta_{kj}$

$$\psi = \sum_k a_k \psi_k, \quad a_k = (\psi_k, \psi) \quad a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix}$$

F' 表象(基矢 ψ'_α) $(\psi'_\alpha, \psi'_\beta) = \delta_{\alpha\beta}$

$$\psi = \sum_\alpha a'_\alpha \psi'_\alpha, \quad a'_\alpha = (\psi'_\alpha, \psi) \quad a' = \begin{pmatrix} a'_1 \\ a'_2 \\ \vdots \end{pmatrix}$$

$$\psi = \sum_{\alpha} a'_{\alpha} \psi'_{\alpha} = \sum_k a_k \psi_k$$

左乘 ψ'_{α} (取标积)

$$a'_{\alpha} = \sum_k (\psi'_{\alpha}, \psi_k) a_k = \sum_k S_{\alpha k} a_k$$

$S_{\alpha k} = (\psi'_{\alpha}, \psi_k)$: F' 表象基矢与 F 表象基矢的标积

矩阵形式

$$\begin{pmatrix} a'_1 \\ a'_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} & \cdots \\ S_{21} & S_{22} & \cdots \\ \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix}$$

$$a' = Sa$$

S : 幺正矩阵

$$SS^+ = S^+S = I \quad (S^+ = S^{-1})$$

● 力学量算符的变换

F 表象

$$L = (L_{kj}) = \begin{pmatrix} L_{11} & L_{12} & \cdots \\ L_{21} & L_{22} & \cdots \\ \vdots & \vdots & \vdots \end{pmatrix}, \quad L_{kj} = (\psi_k, L\psi_j)$$

F' 表象

$$L' = (L'_{\alpha\beta}) = \begin{pmatrix} L'_{11} & L'_{12} & \cdots \\ L'_{21} & L'_{22} & \cdots \\ \vdots & \vdots & \vdots \end{pmatrix}, \quad L'_{\alpha\beta} = (\psi'_\alpha, L\psi'_\beta)$$

$$\psi'_\alpha = \sum_k \psi_k (\psi_k, \psi'_\alpha) = \sum_k S_{\alpha k}^* \psi_k, \quad S_{\alpha k} = (\psi'_\alpha, \psi_k)$$

$$\psi'_\beta = \sum_j \psi_j (\psi_j, \psi'_\beta) = \sum_j S_{\beta j}^* \psi_j$$

$$L'_{\alpha\beta} = \sum_{kj} S_{\alpha k} (\psi_k, L\psi_j) S_{\beta j}^* = \sum_{kj} S_{\alpha k} L_{kj} S_{j\beta}^+ = (SLS^+)_{\alpha\beta}$$

$$L' = SLS^+ = SLS^{-1}$$

- 表象变换不改变算符的本征值(物理要求)

【证明】

在 L 表象中

$$T_L \psi_L = \lambda_L \psi_L$$

在 M 表象中

$$\begin{aligned} T_M \psi_M &= S T_L S^+ S \psi_L \\ &= S T_L S^{-1} S \psi_L \\ &= S T_L \psi_L \\ &= S \lambda_L \psi_L \\ &= \lambda_L S \psi_L \\ &= \lambda_L \psi_M \end{aligned}$$